

Setup

The following code sets up the system by downloading the library from the web.

Make a closed figure function

Here is a list of colors: "blue", "green", "red", "cyan", "magenta", "yellow", "black", "white", "olive", "indigo".

Make a video of drawing a face

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In [34]: # Playing with b

# Create a graph object with equal_axes)
plot = graph_funs(equal_axes=True, hide_axes=False)

# Setup the grid
# Determine the range for X based on reflections and translations.
# Determine the range for Y based on reflections and translations (look at maximum values).
plot.set_XY(minX=-10, maxX=+10, minY=0, maxY=+12)

# Draw different parts of the face and save in different frames

# Frame 1: draw the face:
make_closed_fig(plot, a=2, b=1.4, h=0, k=0, d=3.25, num_of_points=100)
plot.plot_funs(plot_title="Drawing the face", img_name="frame1.png")

# Frame 2: Draw the eyebrows:
make_closed_fig(plot, a=0.5, b=1.1, h=-1.25, k=7.5, d=1, num_of_points=20)
make_closed_fig(plot, a=0.5, b=1.1, h=1.25, k=7.5, d=1, num_of_points=20)
plot.plot_funs(plot_title="Drawing the eyebrows", img_name="frame2.png")

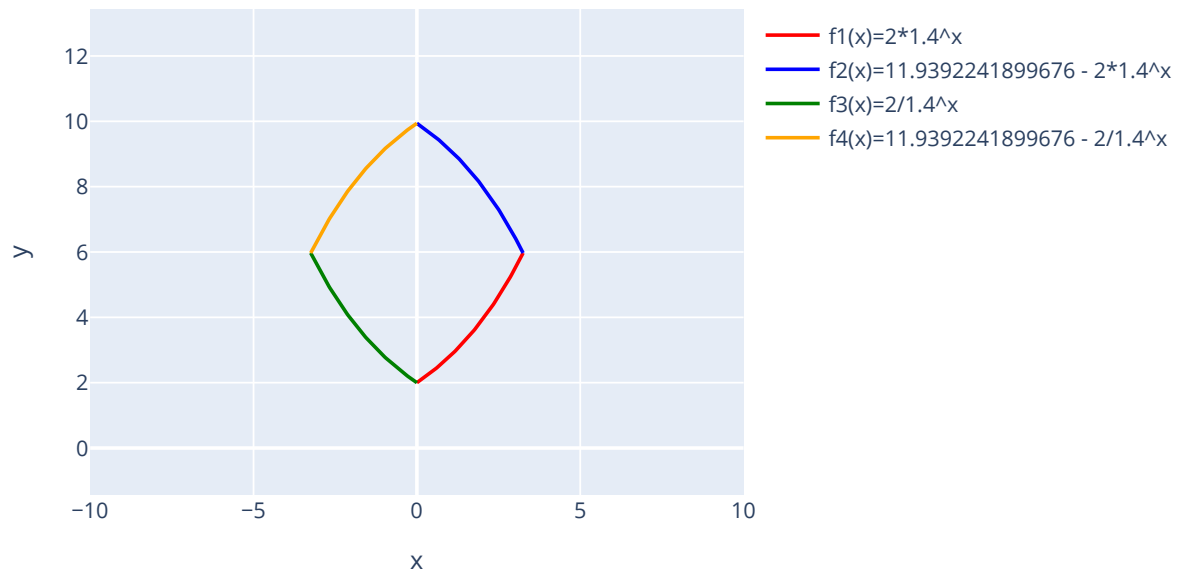
# Frame 3: Draw the eyes:
make_closed_fig(plot, a=1.3, b=1.1, h=-1.25, k=6.2, d=1.1, num_of_points=100)
make_closed_fig(plot, a=1.3, b=1.1, h=1.25, k=6.2, d=1.1, num_of_points=100)
plot.plot_funs(plot_title="Drawing the eye", img_name="frame3.png")

# Frame 4: Draw the mouth:
make_closed_fig(plot, a=0.5, b=1.1, h=0, k=4, d=1.5, num_of_points=20)
plot.plot_funs(plot_title="Drawing the mouth", img_name="frame4.png")

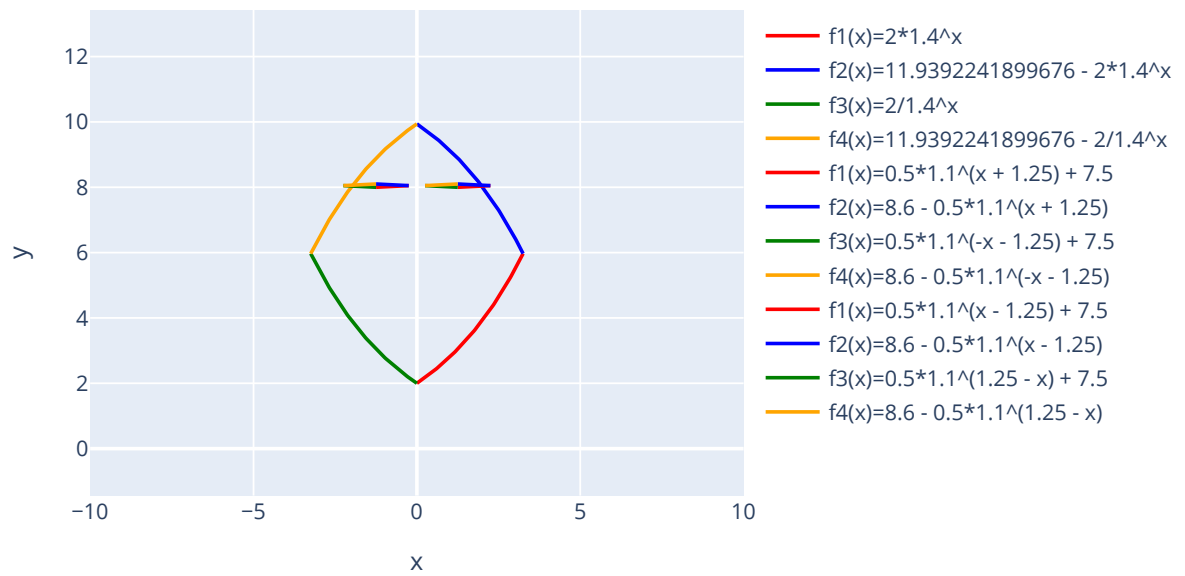
# Frame 5: Draw the nose:
make_closed_fig(plot, a=0.5, b=2, h=0, k=5, d=0.5, num_of_points=20)
plot.plot_funs(plot_title="Drawing the nose", img_name="frame5.png")

# Frame 6: Draw the ears:
make_closed_fig(plot, a=1, b=3, h=-3.17, k=5.5, d=.4, num_of_points=100)
make_closed_fig(plot, a=1, b=3, h=3.17, k=5.5, d=.4, num_of_points=100)
plot.plot_funs(plot_title="Drawing the ears", img_name="frame6.png")
```

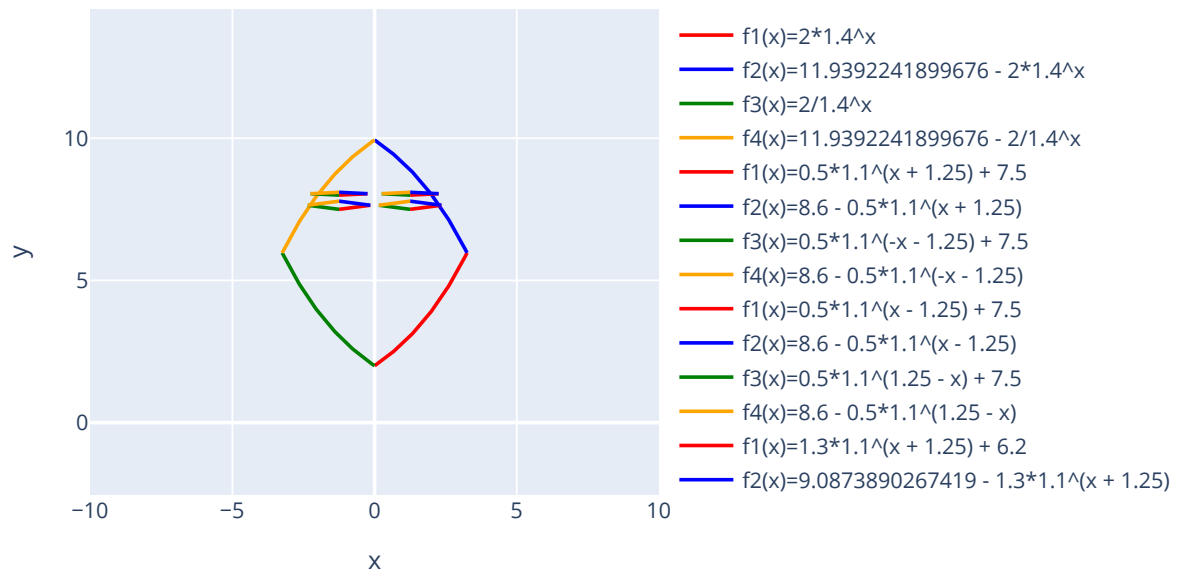
Drawing the face



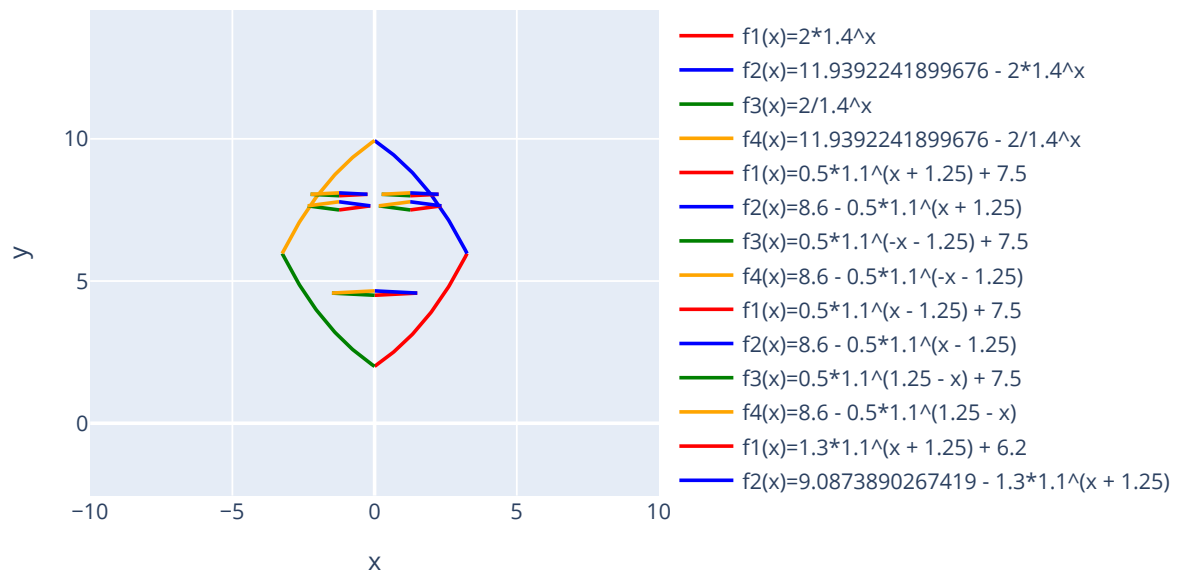
Drawing the eyebrows



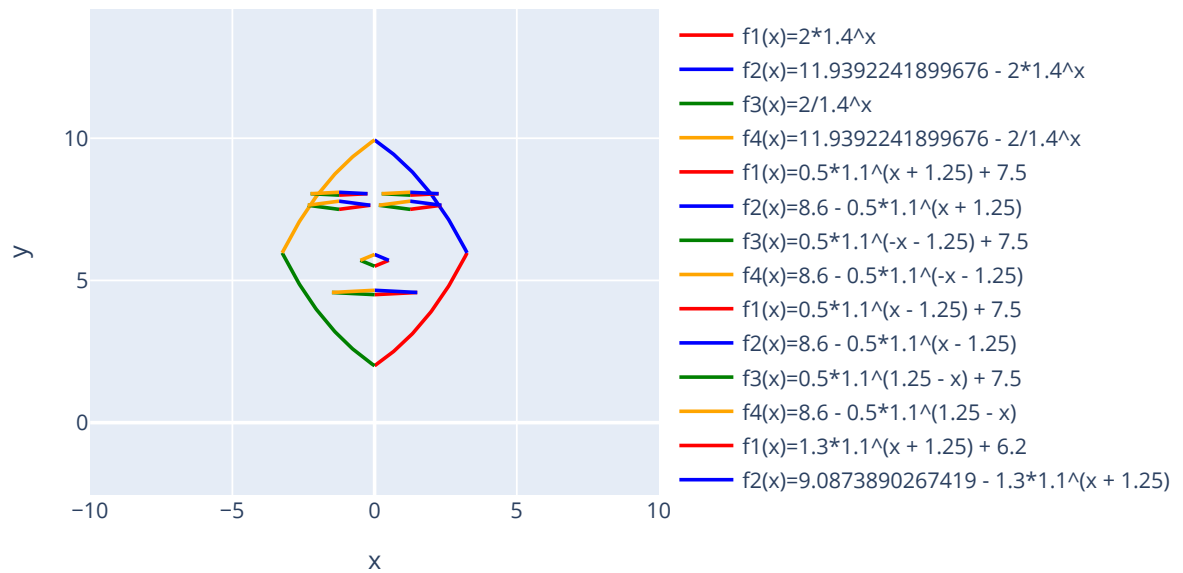
Drawing the eye



Drawing the mouth



Drawing the nose



Drawing the ears

